**THE UNITED REPUBLIC OF TANZANIA**
**ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS-TANZANIA (OHONGSS-T)**
**MWANZA REGION**

**FORM TWO MOCK ASSESSMENT-MAY, 2025**

**041 BASIC MATHEMATICS**

**MARKING SCHEME**

## 1. Numbers and Percentages

### (a) Greatest Common Factor (GCF)

**Problem:** Given the numbers 12, 16, and 20.

**Solution:**

Factors of 12 are: 1,2,3,**4**,6, and 12……. **(01 Mark)**

Factors of 16 are: 1,2,**4**,8, and 16. ……**(01 Mark)**

Factors of 20 are: 1,2,**4**,5,10, and 20……..**(01 Mark)**

From the number factors above, the Greatest Common Factor of the numbers is 4.

**∴The GCF of 12, 16, and 20 is 4 (by Listing method)** …….**(02 Marks)**

**(b)** Given:

Number of boys = 80

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Number of girls = 50

Percentage of boys who were absent = 10%

$\Longrightarrow$Percentage of girls who were absent = 12%

**Solution:**

**Number of boys absent:** 10% of 80=$\frac{10}{100}$ ×80 = 8 boys were absent. ….**(01 Mark)**

**Number of girls absent:** 12% of 50 =$\frac{12}{100}$×50 = 6 girls were absent. …..**(01 Mark)**

**Then, Total number of students:** Total students =Boys + Girls = 80+50 = 130. **….(01 Mark)**

**Total number of absentees:** Number of absentees = 8 + 6 = 14. ….. **(01 Marks)**

⇒**Number of students attended:** Number of students attended =

Total students - Number of absentees Number of students attended =130 – 14 = 116. ….. **(01 Mark)**

**Therefore, 116 students attended the school on that day.**

**2. Ratios and Standard Notation**

**(a) Petrol Consumption**

**Problem:** Given: The ratio of volume of petrol : distance = 4 litre : 80 km.

$\Longrightarrow \begin{matrix} 4litres \equiv 80kilometres \\ xlitres \equiv 320\ kilometres \end{matrix}$ ……..**(01 Mark)**

By cross multiplication;

$x = \frac{4litres \times 320kilometres}{80\ kilometres}$ ……..**(01 Mark)**

$x = 16\ litres$ ……..**(01 Mark)**

Then, converting litres into cubic centimetres ($cm^3$);

$\begin{matrix} 1l = 1000cm^3 \\ 16l = ycm^3 \end{matrix}$ ……..**(01 Mark)**

$\Rightarrow y = \frac{16l \times 1000cm^3}{1l} = 16000cm^3$ (by crossing multiplication)

$\therefore 16000cm^3$ of petrol can be used in a trip of 320 kilometres. ……..**(01 Mark)**

**(b) Population Rounding and Standard Notation**

Given the number of people (residents) =147,500.

(i)**Roundinding**:147,500≈150,000 people correct to the nearest ten thousands. ……..**($02\frac{1}{2}$ Mark)**

∴ To the nearest ten thousand, it is approximately to 150,000 people.

(ii) **Standard notation:** From 147,500 villagers: $147{,}500 = 1.475\times10^5$. (in standard notation; $A\times10^n$ $(1\leq A<10)$

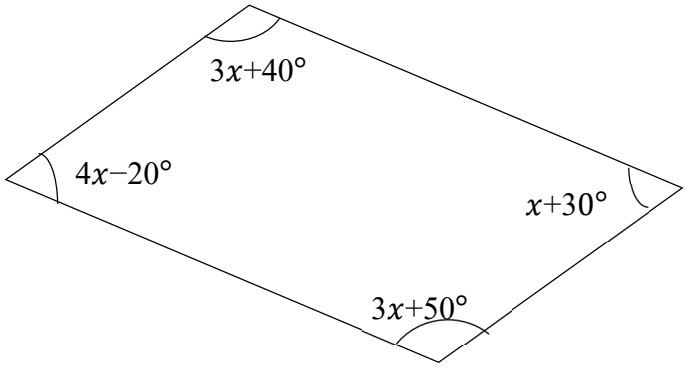
In standard notation, it is $1.475\times10^5$ villagers. ……..**($02\frac{1}{2}$ Marks)**

**3. Geometry**

**(a) Angles of a Quadrilateral**

Given the angles of a quadrilateral as $(4x-20°)$, $(3x+40°)$, $(3x+50°)$, and $(x+30°)$. Find the value of $x$.

**Solution:** The sum of degrees in any quadrilateral polygon is 360∘. Therefore: $(4x-20°)+(3x+40°)+(3x+50°)+(x+30°)=360°$




⇒The sum of degrees in any quadrilateral polygon gives 360° ……..**(01 Mark)**

⇒ $(4x-20°)+(3x+40°)+(3x+50°)+(x+30°) = 360°$

$(4x + 3x + 3x + x) + (40°-20° + 50° + 30°) = 360°$……..**(02 Marks)**

$11\,x + 100° = 360°$ (By collecting like term)

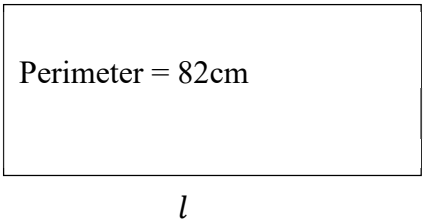
$11\,x = 360° - 100°$ ……..**(01 Mark)**

$11\,x = 260° \;\Rightarrow\; x = \frac{260°}{11} = 23.6°$

∴ The vaule of $x$ is $23.6°$ ……..**(01 Mark)**

**(b) Area of a Rectangle**

Let $l$ represents the length of a rectangle and $w$ represents the width of the same rectangle. Given: $\Rightarrow l - 3w = 5$




……..**(00$\frac{1}{2}$ Marks)**

From perimeter of a rectangle, P = 2($l$ + $w$). ……..**(00$\frac{1}{2}$ Marks)**

P = 2($l$ + $w$) = 82cm

Since $l - 3w = 5$ , $l$ = 5 +3$w$ ……..**(00$\frac{1}{2}$ Marks)**

Substituting $l$ = 5 +3$w$ into the equation 2($l$ + $w$) = 82,

2(5 + 3$w$ + $w$) = 82cm……..**(00$\frac{1}{2}$ Marks)**

2(5 + 4$w$) = 82cm $\Rightarrow 5 + 4w = 41cm$

$4w = 41 - 5$

$4w = 36$

$\frac{4w}{4} = \frac{36}{4}$ , $w = 9cm$……..**(01 Mark)**

If w = 9cm, then $l$ = 5 +3$w$ = 5 +3(9)

$l$ = 5 + 27 = 32cm

The Length (l) = 32cm, width (w) = 9cm

Then the area (A) of a rectangle is given by A= $l$ × w ……..**(00$\frac{1}{2}$ Marks)**

A =32 cm× 9 cm A=288 cm$^2$

∴The area of this rectangle is 288 cm$^2$. ……..**(00$\frac{1}{2}$ Marks)**

**4. Simultaneous Equations**

**(a) Forming a System of Equations**

Let m represent the cost of a box of marker pens and n represent the cost of a box of correction fluid. ……..**(01 Mark)**

The given information are summarized as follows

From the first order: 50m + 20n = 40,000

5m + 2n = 4000 … (i) (after simplification) ……..**(01 Mark)**

From the second order 10m + 20n = 24,000

m + 2n = 2,400 … (ii) (after simplification) ……..**(01 Mark)**

Therefore, a system of simuljteneous equation is given below:

$$\begin{cases} 5m + 2n = 4{,}000 \\ m + 2n = 2.400 \end{cases}$$ ……..**(01 Mark)**

**(b) Solving the System of Equations**

To get the cost of a box marker pens (m) and that of a box of correction fluid, we solve the system of simultaneously equation in (a)

$$\begin{cases} 5m + 2n = 4{,}000 \ldots (i) \\ m + 2n = 2.400 \ldots (ii) \end{cases}$$

Using combination method (elimination and substitution method) ……..**(01 Mark)**

Eliminating n to get m

$$-\begin{cases} 5m + 2n = 4{,}000 \\ m + 2n = 2.400 \end{cases}$$ ……..**(01 Mark)**

$$5m - m = 4{,}000 - 2{,}400$$

$$4m = 1{,}600$$

$$\frac{4m}{4} = \frac{1{,}600}{4}, m = 400 \text{ Tshs}$$ ……..**(01 Mark)**

Then, from equation (i)

$$5m + 2n = 4{,}000$$

Substituting $m = 400$

$5(400) + 2n = 4{,}000$ ……..**(01 Mark)**

$2{,}000 + 2n = 4{,}000$

$2n = 4{,}000 - 2{,}000$

$2n = 2{,}000 \Rightarrow \frac{2n}{2} = \frac{2000}{2}, n = 1000$ Tshs ……..**(01 Mark)**

The cost of a box of marker pens is 400 Tanzanian shillings while that of a box of correction fluid is 1,000 Tanzanian shillings. ……..**(01 Mark)**

**5. Ratios and Simple Interest**

**(a) Sharing Counter Books**

Given: Number of counter books = 120.

(i) Ratio of Hilda : Moghani : Juvinalis = 2:5:3.

Total ratio =2+5+3=10. ……..**(01 Mark)**

- **Hilda's share:** Share = $\frac{2}{10}$×120=2×12=24 counter books. ……..**(00$\frac{1}{2}$ Marks)**
- **Moghani's share:** Share = $\frac{5}{10}$×120=5×12=60 counter books. ……..**(00$\frac{1}{2}$ Marks)**
- **Juvinalis' share:** Share=$\frac{3}{10}$×120=3×12=36 counter books. ……..**(00$\frac{1}{2}$ Marks)**

**(ii) Difference between largest and smallest share:**

Largest share = Moghani's share = 60 counter books. ……..**(00$\frac{1}{2}$ Marks)**

Smallest share = Hilda's share =24 counter books. ……..**(00$\frac{1}{2}$ Marks)**

⇒Largest share - Smallest share =60−24=36 counter books. ……..**(01 Mark)**

∴The difference between the largest and the smallest share is 36 counter books. ……..**(00$\frac{1}{2}$ Marks)**

**(b) Simple Interest Calculation**

Given:

- Time of investment (T) = 3 years.
- Simple interest (I) =8 7,750 Tsh. ……..**(01 Mark)**
- Interest rate (R) = 4.5%.

Principal (P) amount. Required.

The formula for Simple Interest is $I = \frac{P \times R \times T}{100}$……..**(01 Mark)**

Substitute the given values into the formula: $87{,}750Tsh = \frac{P \times 4.5 \times 3}{100}$ ……..**(01 Mark)**

Multiply both sides by 100: $87{,}750 \times 100 = P \times 4.5 \times 3 \ = P \times 13.5$

Divide by 13.5 to find P: $P = \frac{775{,}000Tsh}{13.58}$ , P = 650,000 Tsh. ……..**(01 Mark)**

∴A trader deposited 650,000 Tanzanian shillings as a principal amount...**(01 Mark)**

## 6. Coordinates and Transformations

### (a) Gradient of a Line

Given:

- Gradient (m)=4
- Points A(−1,2r) and B(3,4)

From, gradient(m) $= \frac{change\ in\ y\ coordinates}{change\ in\ x\ coordinates}$ …..**(00$\frac{1}{2}$ Marks)**

$m = \frac{y_2 - y_1}{x_2 - x_1}$ …..**(00$\frac{1}{2}$ Marks)**

$(x_1, y_1 = (-1, 2r)\ and\ x_2, y_2 = (3,4)$

$\Rightarrow 4 = \frac{4-2r}{3-(-1)}$

$\Rightarrow 4 = \frac{4-2r}{3+1} \ \Rightarrow 4 = \frac{4-2r}{4}$ ……..**(01 Mark)**

$\Rightarrow 16 = 4 - 2r$

$-2r = 16 - 4$

$-2r = 12, \Rightarrow r = -\frac{12}{2} = -6$ ……..**(01 Mark)**

∴The value of $r$ is −6. ……..**(01 Mark)**

**(b) Enlargement of a Quadrilateral**

Given vertices O(0,0), A(1,0), B(1,1), C(0,1). Enlargement factor (k)=4.

From $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}(k)$ …..**(00$\frac{1}{2}$ Marks)**

$\Rightarrow \quad \boldsymbol{O'} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}(4) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = (0,0)$ …..**(00$\frac{1}{2}$ Marks)**

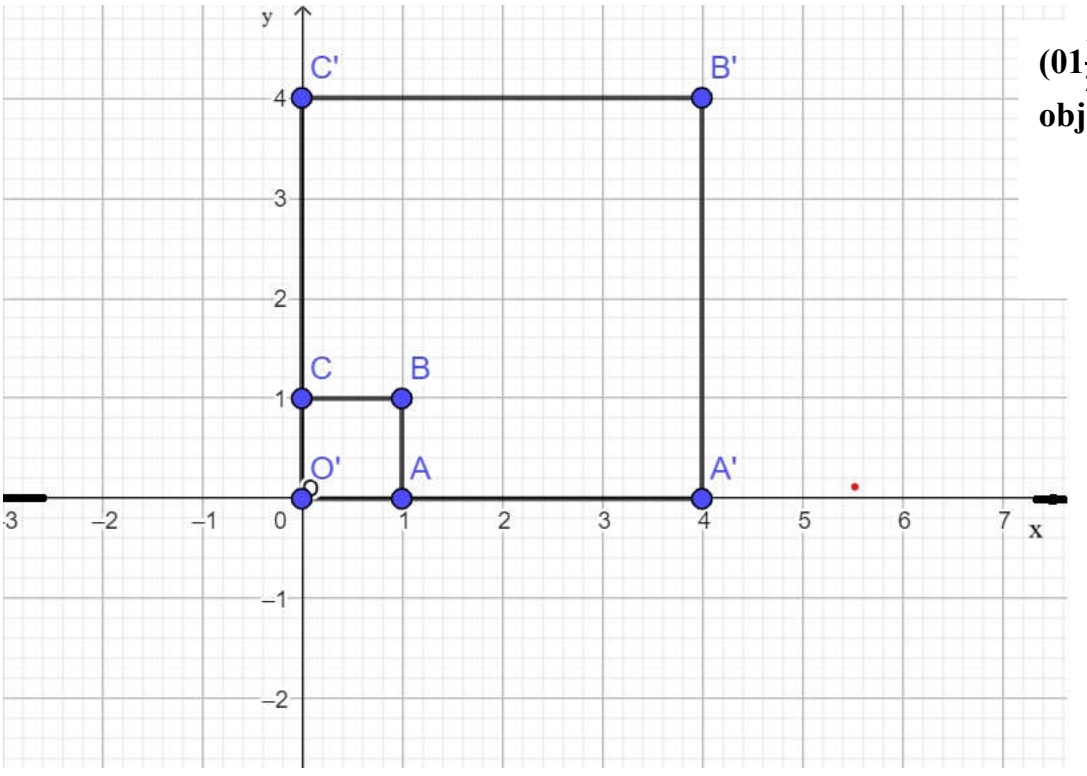
$\boldsymbol{A'} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}(4) = \begin{pmatrix} 04 \\ 0 \end{pmatrix} = (4,0)$ …..**(00$\frac{1}{2}$ Marks)**

$\boldsymbol{B'} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}(4) = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = (4,4)$ …..**(00$\frac{1}{2}$ Marks)**

$\boldsymbol{C'} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}(4) = \begin{pmatrix} 0 \\ 4 \end{pmatrix} = (0,4)$ …..**(00$\frac{1}{2}$ Marks)**

∴The images are $O'(0,0), A'(4,0), B'(4,4), and\ C'(0,4)$. …..**(00$\frac{1}{2}$ Marks)**

**(b) Graphical Representation**




**(01$\frac{1}{2}$ @ = 03 Marks) object and Image.**

**7. Exponents and Logarithms**

**(a)** (i) Given: $2^{x=2} \times 3^{2y+3} = 36$

Express 36 as a product of prime factors gives $36 = 2^2 \times 3^2$…..**(00$\frac{1}{2}$ Marks)**

$\Rightarrow 2^{x+2} \times 3^{2y+3} = 2^2 \times 3^2$…..**(00$\frac{1}{2}$ Marks)**

Then, comparing bases and exponents,

$2^{x+2} = 2^2$ and $3^{2y+3} = 3^2$……..**(01 Mark)**

$\Rightarrow x + 2 = 2$ and $2y + 3 = 2$ (equal bases implies equal exponets)

$x = 2 - 2 = 0$ and $2y = 2 - 3 = -1$

$x = 0$ and $2y = -1, y = -\frac{1}{2}$ …..**(00$\frac{1}{2}$ Marks)**

∴The value of $x = 0$ and $y = -\frac{1}{2}$

(ii) Required to simplify $\sqrt{45} \times \sqrt{12}$

$\sqrt{45} \times \sqrt{12} = \sqrt{9 \times 5} \times \sqrt{3 \times 4}$ …..**(01 Mark)**

$= \left(\sqrt{9} \times \sqrt{5}\right) \times \left(\sqrt{4} \times \sqrt{3}\right)$

$= 3\sqrt{5} \times 2\sqrt{3}$…..**(01 Mark)**

$= 6\sqrt{15}$

$\therefore \sqrt{45} \times \sqrt{12} = 6\sqrt{15}$ …..**(00$\frac{1}{2}$ Marks)**

**(b)** Given: $\log_3(x + 6) - \log_3 x = 2$

Applying the laws of logarithms;

$\log_a p - \log_a q = \log_a \left(\frac{p}{q}\right)$…..**(00$\frac{1}{2}$ Marks)**

$\Rightarrow \log_3 \left(\frac{x+6}{x}\right) = 2$ …..**(01 Mark)**

Then writing in exponential form;

If $\log_b a = x$ then $a = b^x$ …..**(00$\frac{1}{2}$ Marks)**

$\frac{x+6}{x} = 9$

$x + 6 = 9x$ …..**(00$\frac{1}{2}$ Marks)**

$\Rightarrow 6 = 9x - x$

$6 = 8x$ …..**(00$\frac{1}{2}$ Marks)**

$x = \frac{6}{8} = \frac{3}{4}$ …..**(01 Mark)**

∴ The value of $x = \frac{3}{4}$ …..**(00$\frac{1}{2}$ Marks)**

## 8. Congruence and Similarity of Triangles

### (a) Proving Triangle Congruence

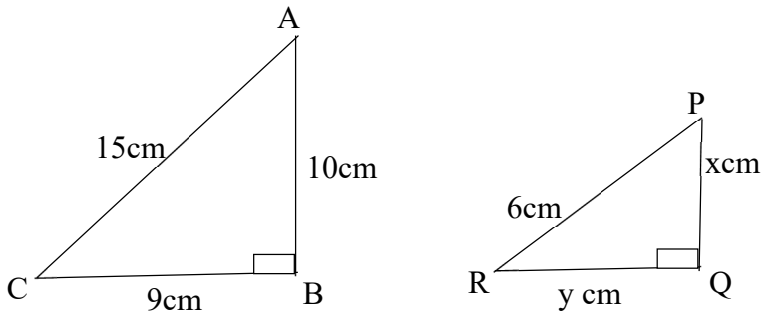
From the given figure AC = CB and D$\hat{A}$C = D$\hat{B}$C = 90°. You are required to prove that △ACD≅△CBD.

T △ACD and △CBD indicate that : ..**(01 Mark**

1. **Side:** AC=CB (Given). …..**(01 Mark)**
2. **Angle:** D$\hat{A}$C = D$\hat{B}$C = 90° (Given). …..**(01 Mark)**
3. **Side:** CD is a common side to both triangles. …..**(01 Mark)**

Therefore, △ACD≅△CBD by the SAS (Side-Angle-Side) postulate. …..**(01 Mark)**

**(b) Similar Triangles and Side Lengths**




Consider two similar triangles △ABC with sides AB=10 cm, AC=15 cm, BC=9 cm. △PQR with sides PQ=x cm, PR=6 cm, QR=y cm. Given that △ABC∼△PQR. Find the lengths of the other two sides (x and y).

Since △ABC∼△PQR, it follows that

$\frac{\overline{BC}}{\overline{QR}} = \frac{\overline{AB}}{\overline{PQ}} = \frac{\overline{AC}}{\overline{PR}}$ …..**(01 Mark)**

Consider $\frac{\overline{AB}}{\overline{PQ}} = \frac{\overline{AC}}{\overline{PR}}$

$\overline{AB} = 10cm, \overline{PQ} = xcm, \overline{AC} = 15cm, PR = 6cm$, $\overline{BC}$=9cm and $QR = ycm$

$$\Rightarrow \frac{10cm}{xcm} = \frac{15cm}{6cm}$$

$$x = \frac{10cm \times 6cm}{15cm} = 4cm$$ …..**(01 Mark)**

$$Also, \frac{\overline{BC}}{\overline{QR}} = \frac{\overline{AC}}{\overline{PR}}$$

$$\frac{9cm}{ycm} = \frac{15cm}{6cm}, y = \frac{9cm \times 6cm}{15cm} = 3.6cm$$ …..**(01 Mark)**

∴The lengths of the other two sides are 3.6 cm and 4 cm respectively.

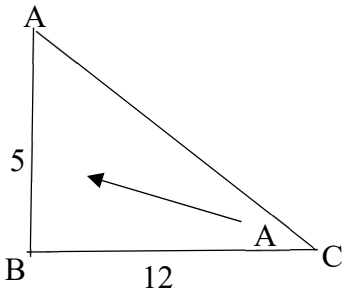
**9. Trigonometry and Pythagoras Theorem**

**(a) Trigonometric Expression Evaluation**

Given: 12tanA =5.

⇒ tanA= $\frac{5}{12}$ …..**(00$\frac{1}{2}$ Marks)**

From $\tan A = \frac{opposite}{adjacent} = \frac{5}{12}$ …..**(00$\frac{1}{2}$ Marks)**




From Pythagoras theorem,

$\overline{AB}^2 + \overline{BC}^2 = \overline{AC}^2$

$5^2 + 12^2 = \overline{AC}^2$

$25 + 144 = \overline{AC}^2$ …..**(00$\frac{1}{2}$ Marks)**

$\overline{AC}^2 = 169$

$\overline{AC} = \sqrt{169} = 13$

$AC = 13$…..**(00$\frac{1}{2}$ Marks)**

Then, $\sin A = \frac{Opposite}{Hypotenuse} = \frac{\overline{AB}}{\overline{AC}} = \frac{5}{13}$

$\sin A = \frac{5}{13}$ …..**(00$\frac{1}{2}$ Marks)**

$\cos A = \frac{Adjacent}{Hyponetuse} = \frac{\overline{BC}}{\overline{AC}} = \frac{12}{13}$

$\cos A = \frac{12}{13}$ …..**(00$\frac{1}{2}$ Marks)**

$$\Rightarrow \frac{\sin A - 2\cos A}{1+\sin A} = \frac{\frac{5}{13} - 2(\frac{12}{13})}{1+\frac{5}{13}}$$

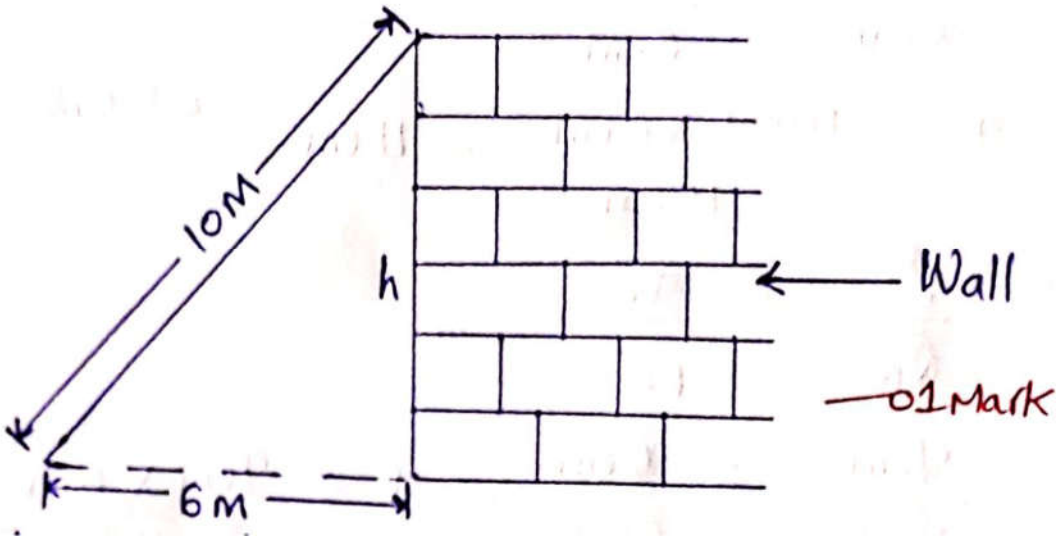
$$= \frac{\frac{5}{13} - \frac{24}{13}}{1+\frac{5}{13}}$$

$= \frac{-19}{13} \div \frac{18}{13} = \frac{-19}{13} \times \frac{13}{18}$ …..**(01 Mark)**

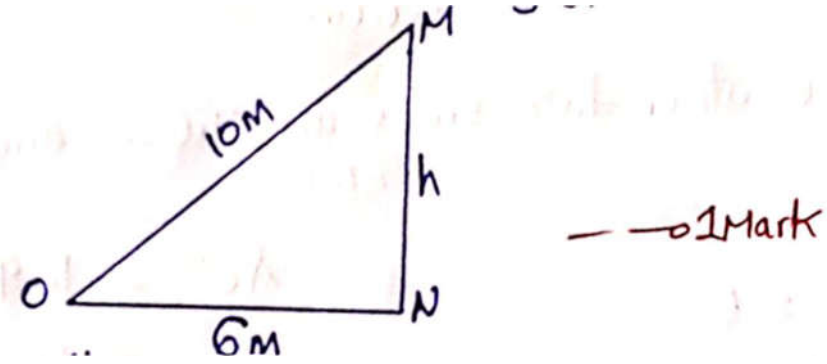
$= \frac{-19}{18}$

∴The value of $\frac{\sin A - 2\cos A}{1+\sin A} = \frac{-19}{18}$ …..**(01 Mark)**

**(b) Ladder Problem (Pythagoras Theorem)**




**Consider a right-angled triangle**




Apply Pythagoras theorem to $\Delta OMN$

$\overline{ON}^2 + \overline{NM}^2 = \overline{OM}^2$

$6^2 + h^2 = \overline{10}^2$

$36m^2 + h^2 + 100m^2$

$h^2 = 100m^2 - 36m^2$

$h = 64m^2$

$h = \sqrt{64m^2} = 8m$…..**(01 Mark)**

∴ The top of the ladder should reach the height of $8m$…..**(01 Mark)**

**10. Sets and Statistics**

**(a) Set Theory**

Let G represent students taking Geology and E represents students taking Education. Given:

- Total students μ = n(G∪E) + n(G∪E)′ = 40 …..**(00$\frac{1}{2}$ Marks)**
- n(G) = 18
- n(G∩E) = 15
- n(G∪E)′ = 8 (students taking neither Geology nor Education) …..**(00$\frac{1}{2}$ Marks)**

From the total students formula: μ = n(G∪E) + n(G∪E)′

40 = n(G∪E) +8

n(G∪E) = 40−8 = 32. …..**(01 Mark)**

Then (G∪E) = n(G) + n(E) − n(G∩E) …..**(01 Mark)**

32 = 18+ n(E) − 15

32 =3 + n(E) …..**(01 Mark)**

n(E) = 32−3

n(E)=29.

∴ 29 students take Education. …..**(00$\frac{1}{2}$ Marks)**

**(b) Frequency Distribution and Modal Class**

(i) **Total number of people:** Total number of people = Sum of all frequencies =30+45+25+55+80+40=275.

There are 275 people at Picha ya Ndege village. …..**(01 Mark)**

(ii) **Cumulative frequency table:**

| Mass Range (Kg) | Frequency | Cumulative Frequency |
|---|---|---|
| 0-9 | 30 | 30 |
| 10-19 | 45 | 30+45=75 |
| 20-29 | 25 | 75+25=100 |
| 30-39 | 55 | 100+55=155 |
| 40-49 | 80 | 155+80=235 |
| 50-59 | 40 | 235+40=275 |

…..**(02 Mark)**

(iii) **Modal class and class mark:** The modal class is the one with the highest frequency. From the table, the highest frequency is 80, which corresponds to the mass range 40−49. So, the **modal class is 40−49.** …..**(00$\frac{1}{2}$ Marks)**

The class mark (X) is given by:

$Class\ mark\ (X) = \frac{Lower\ limit + Upper\ limit}{2}$ …..**(00$\frac{1}{2}$ Marks)**

$= \frac{40-49}{2}$:

$Class\ mark \quad = \frac{89}{2} = 44.5$ …..**(00$\frac{1}{2}$ Marks)**

The class mark for the modal class is 44.5 …..**(00$\frac{1}{2}$ Marks)**